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Electromechanical brake system and method of controlling the same

Abstract

The present disclosure relates to an electromechanical brake system and a method of controlling the same, and according to one aspect of the present disclosure, the method of controlling the electromechanical brake system may include calculating a required fluid amount according to an actual pressure required for brake oil required to generate a braking force by the motor, calculating a ratio of the calculated required fluid amount and a required fluid amount reference according to a map, and applying the required fluid amount as the required fluid amount reference according to the calculated ratio.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0003532, filed on Jan. 10, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field

(2) Embodiments of the present disclosure relate to an electromechanical brake system and a method of controlling the same, and more specifically, to an electromechanical brake system capable of controlling a required fluid amount of brake oil and a method of controlling the same.

2. Description of the Related Art

(3) A vehicle necessarily includes a brake system for braking. Recently, such a brake system electronically controls braking hydraulic pressure transmitted toward a wheel cylinder mounted on a wheel in order to obtain a stronger and more stable braking force.

(4) The conventional brake systems use a method of supplying hydraulic pressure required for braking to a wheel cylinder using a mechanically connected booster when a driver steps on a brake pedal. However, as the demand for effectively implementing a braking function in various environments by minutely responding to the operating environment of a vehicle increases, an electromechanical brake system is used. The electromechanical brake system uses a hydraulic supply device for converting the driver's braking intention into a signal using a pedal displacement sensor for detecting the displacement of a brake pedal when the driver steps on the brake pedal and supplying hydraulic pressure required for braking to a wheel cylinder according to the signal.

(5) Since a braking force is generated by a motor in such an electromechanical brake system, a reference amount is needed for the required fluid amount of brake oil. The required fluid amount of the brake oil represents the relationship between a position of the motor and the hydraulic pressure of the brake oil.

(6) However, since the required fluid amount of the brake oil varies depending on a state of a vehicle, there is a problem in that when the required fluid amount of the brake oil is significantly changed as compared to the reference amount, brake control performance is degraded.

SUMMARY

(7) Therefore, it is an aspect of the present disclosure to provide an electromechanical brake system that is applicable when a required amount of brake oil is changed as compared to a reference amount and a method of controlling the same.

(8) Additional aspects of the disclosure will be set forth in part in the description which follows and, in part, will be obvious from the detailed description, or may be learned by practice of the disclosure.

(9) In accordance with one aspect of the present disclosure, a method of controlling an electromechanical brake system includes calculating a required fluid amount according to an actual pressure required for brake oil required to generate a braking force by a motor, calculating a ratio of the calculated required fluid amount and a required fluid amount reference according to a map, and applying the required fluid amount as the required fluid amount reference according to the calculated ratio.

(10) The method may further include determining whether to monitor the required fluid amount, wherein the determining of whether to monitor the required fluid amount comprises performing the monitoring the required fluid amount in cases in which the required fluid amount is continuously changed.

(11) The cases in which the required fluid amount is continuously changed may be cases excluding one or more of a case in which an estimated brake temperature is higher than a set temperature and a case in which an outside air temperature is lower than the set temperature.

(12) The cases in which the required fluid amount is continuously changed may be cases excluding one or more among a case in which the required fluid amount is actively increased by a driver, a case in which vehicle suddenly decelerates, and a case in which a brake oil pressure is low.

(13) The method may further include determining whether the required fluid amount is applicable as the required fluid amount reference, wherein the applying of the required fluid amount as the required fluid amount reference may be performed when it is determined that required fluid amount is applicable in the determining of whether the required fluid amount is applicable.

(14) In the determining of whether the required fluid amount is applicable comprises determining

that the required fluid amount is applicable when the ratio of the calculated required fluid amount and the required fluid amount reference is calculated in a plurality of sections divided according to pressure.

(15) In the determining of whether the required fluid amount is applicable comprises determining that the required fluid amount is applicable when the ratio of the required fluid amount and the required fluid amount reference is calculated multiple times in one or more sections among a plurality of sections divided according to pressure.

(16) In the determining of whether the required fluid amount is applicable comprises determining that the required fluid amount is applicable when a traveling distance of a vehicle to which the braking force is provided by the motor is a predetermined distance or more.

(17) In the determining of whether the required fluid amount is applicable comprises determining that the required fluid amount is applicable when a traveling distance of a vehicle to which the braking force is provided by the motor is a predetermined distance or more after the required fluid amount has been applied as the required fluid amount reference.

(18) The ratio of the required fluid amount and the required fluid amount reference may be calculated by dividing the required fluid amount by the required fluid amount reference according to pressure.

(19) Meanwhile, according to one aspect of the present disclosure, an electromechanical brake system includes a temperature sensor for measuring one or more of a brake oil temperature and an outside air temperature in order to monitor a required fluid amount according to an actual pressure required for brake oil required to generate a braking force by the motor, a pressure sensor for measuring a pressure for the required fluid amount, and a controller for determining whether to monitor the required fluid amount based on information measured by the temperature sensor and the pressure sensor and calculate a ratio of the required fluid amount and a required fluid amount reference according to a map and the required fluid amount.

(20) The controller may apply the required fluid amount as the required fluid amount reference according to the ratio of the required fluid amount and the required fluid amount reference according to the map.

(21) The controller may monitor the required fluid amount in cases in which the required fluid amount is continuously changed.

(22) The cases in which the required fluid amount is continuously changed may be cases excluding one or more of a case in which an estimated brake temperature is higher than a set temperature and a case in which an outside air temperature is lower than the set temperature.

(23) The cases in which the required fluid amount is continuously changed may be cases excluding one or more among a case in which the required fluid amount is actively increased by a driver, a case in which a vehicle suddenly decelerates, and a case in which a brake oil pressure is low.

(24) The controller may determine whether the required fluid amount is applicable as the required fluid amount reference and apply the required fluid amount as the required fluid amount reference.

(25) The controller may determine that the ratio of the calculated required fluid amount and the required fluid amount reference is applicable when calculated in a plurality of sections divided according to the pressure.

(26) The controller may determine that the ratio of the required fluid amount and the required fluid amount reference is applicable when calculated multiple times in one or more among a plurality of sections divided according to the pressure.

(27) The controller may determine that the ratio of the required fluid amount and the required fluid amount reference is applicable when a traveling distance of a vehicle to which the braking force is provided by the motor is a predetermined distance or more.

(28) The controller may determine that the ratio is applicable when a traveling distance of a vehicle to which the braking force is provided by the motor is a predetermined distance or more after the required fluid amount has been applied as the required fluid amount reference.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and/or other aspects of the disclosure will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings of which:
- (2) FIG. 1 is a hydraulic circuit diagram showing an electromechanical brake system according to one embodiment of the present disclosure;
- (3) FIG. 2 is a block diagram showing the electromechanical brake system according to one embodiment of the present disclosure;
- (4) FIG. 3 is a graph for describing determination of a section in which monitoring is possible in the electromechanical brake system according to one embodiment of the present disclosure;
- (5) FIG. 4 is a graph for describing calculation of an increase rate of a required fluid amount in the electromechanical brake system according to one embodiment of the present disclosure; and
- (6) FIG. 5 is a flowchart for describing a method of controlling an electromechanical brake system according to one embodiment of the present disclosure.

DETAILED DESCRIPTION

- (7) The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. Accordingly, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be suggested to those of ordinary skill in the art. The progression of processing operations described is an example; however, the sequence of and/or operations is not limited to that set forth herein and may be changed as is known in the art, with the exception of operations necessarily occurring in a particular order. In addition, respective descriptions of well-known functions and constructions may be omitted for increased clarity and conciseness.
- (8) Additionally, exemplary embodiments will now be described more fully hereinafter with reference to the accompanying drawings. The exemplary embodiments may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein. These embodiments are provided so that this disclosure will be thorough and complete and will fully convey the exemplary embodiments to those of ordinary skill in the art. Like numerals denote like elements throughout.
- (9) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. As used herein, the term “and/or,” includes any and all combinations of one or more of the associated listed items.
- (10) It will be understood that when an element is referred to as being “connected,” or “coupled,” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected,” or “directly coupled,” to another element, there are no intervening elements present.
- (11) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the,” are intended to include the plural forms as well, unless the context clearly indicates otherwise.
- (12) Reference will now be made in detail to the exemplary embodiments of the present disclosure, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout.
- (13) The expression, “at least one of a, b, and c,” should be understood as including only a, only b, only c, both a and b, both a and c, both b and c, or all of a, b, and c.
- (14) Referring to FIG. 1, an electromechanical brake system 1 according to one embodiment of the present disclosure will be described.

(15) The electromechanical brake system **1** generally includes a master cylinder **20** for generating hydraulic pressure, a reservoir **30** coupled to an upper portion of the master cylinder **20** to store a brake fluid, an input rod **12** for pressing the master cylinder **20** according to a force pressing a brake pedal **10**, a wheel cylinder **40** for receiving the hydraulic pressure and performing braking of respective vehicle wheels RR, RL, FR, and FL, a pedal position sensor **11** for detecting displacement of the brake pedal **10**, and a simulation device **50** for providing a reaction force according to the force pressing the brake pedal **10**.

(16) In addition, although not shown in FIG. **1**, a wheel speed sensor may be provided in each wheel cylinder **40** to measure a speed of each of the wheels FL, RR, RL, and FR.

(17) The master cylinder **20** may have at least one chamber to generate the hydraulic pressure. For example, the master cylinder **20** may have two chambers, each chamber may be provided with a first piston **21a** and a second piston **22a**, and the first piston **21a** may be connected to the input rod **12**. In addition, the master cylinder **20** may have first and second hydraulic ports **24a** and **24b** through which hydraulic pressure is discharged from each of the two chambers.

(18) Meanwhile, the master cylinder **20** may have two chambers, thereby securing safety upon failure. For example, one of the two chambers may be connected to the front right wheel FR and the rear left wheel RL of the vehicle, and the other chamber may be connected to the front left wheel FL and the rear right wheel RR. As described above, the two chambers may be independently configured so that the braking of a vehicle is possible even when one chamber fails.

(19) In addition, a first spring **21b** may be provided between the first piston **21a** and the second piston **22a** of the master cylinder **20**, and a second spring **22b** may be provided between the second piston **22a** and an end of the master cylinder **20**.

(20) The first spring **21b** and the second spring **22b** are provided in each of the two chambers, and as the displacement of the brake pedal **10** varies, the first piston **21a** and the second piston **22a** are pushed, and an elastic force is stored in the first spring **21b** and the second spring **22b**. In addition, when a force pushing the first piston **21a** becomes smaller than the elastic force, the first spring **21b** and the second spring **22b** may push the first and second pistons **21a** and **22a** back to original states using the stored elastic force.

(21) Meanwhile, the input rod **12** for pushing the first piston **21a** of the master cylinder **20** may be in close contact with the first piston **21a**. In other words, a gap between the master cylinder **20** and the input rod **12** may not be present. Therefore, when the brake pedal **10** is pressed, the master cylinder **20** may be directly pressed without a pedal invalid stroke section.

(22) The simulation device **50** may be connected to a first backup flow path **251** to be described below to provide a reaction force according to the force pressing the brake pedal **10**. As the reaction force is provided as much as compensating for the pressing force provided by a driver, the driver may minutely adjust a braking force as intended.

(23) The simulation device **50** includes a pedal simulator including a simulation chamber **51** provided to store a brake fluid discharged from the first hydraulic port **24a** of the master cylinder **20**, a reaction force piston **52** provided in the simulation chamber **51**, and a reaction force spring **53** for elastically supporting the reaction force piston **52** and a simulator valve **54** connected to a rear end of the simulation chamber **51**.

(24) The reaction force piston **52** and the reaction force spring **53** are installed to have displacement in a certain range within the simulation chamber **51** by the brake fluid introduced into the simulation chamber **51**.

(25) The simulator valve **54** may be provided on a flow path connecting the rear end of the simulation chamber **51** and the reservoir **30**. A front end of the simulation chamber **51** may be connected to the master cylinder **20**, and the rear end of the simulation chamber **51** may be connected to the reservoir **30** through the simulator valve **54**. Therefore, even when the reaction force piston **52** returns, the brake fluid in the reservoir **30** may be introduced through the simulator valve **54**, and thus the entire inside of the simulation chamber **51** may be filled with the brake fluid.

(26) Meanwhile, several reservoirs **30** are shown in the drawing, and each reservoir **30** uses the same reference numeral. However, the reservoirs may be provided as the same component or different components. For example, the reservoir **30** connected to the simulation device **50** may be the same as the reservoir **30** connected to the master cylinder **20** or may be a storage capable of storing a brake fluid separately from the reservoir **30** connected to the master cylinder **20**.

(27) The simulator valve **54** may be configured as a normally closed type solenoid valve that remains in a normally closed state. The simulator valve **54** may be opened when the driver applies a pressing force to the brake pedal **10** and may transmit the brake fluid in the simulation chamber **51** to the reservoir **30**.

(28) In addition, a simulator check valve **55** may be installed between the simulation chamber **51** and the reservoir **30** to be connected parallel to the simulator valve **54**. The simulator check valve **55** may allow the brake fluid in the reservoir **30** to flow to the simulation chamber **51** and block the brake fluid in the simulation chamber **51** from flowing to the reservoir **30** through a flow path on which the check valve **55** is installed. The brake fluid may be supplied into the simulation chamber **51** through the simulator check valve **55** when the force pressing the brake pedal **10** is released, thereby ensuring the quick return of a pedal simulator pressure.

(29) A hydraulic supply device **100** may include a hydraulic provision unit **110** for providing the hydraulic pressure of the brake fluid transmitted to the wheel cylinder **40**, a motor **120** for generating a rotational force by an electrical signal of the pedal position sensor **11**, and a power converter **130** for converting a rotational motion of the motor **120** into a linear motion and transmitting the linear motion to the hydraulic provision unit **110**. Alternatively, the hydraulic provision unit **110** may also be operated not by a driving force supplied from the motor **120** but by pressure supplied from a high-pressure accumulator.

(30) The electromechanical brake system **1** may include the hydraulic supply device **100** mechanically operated after receiving a driver's braking intention as the electrical signal from the pedal position sensor **11** for detecting the displacement of the brake pedal **10**, a hydraulic control unit **200** including first and second hydraulic circuits **201** and **202**, each of which controls the flow of the hydraulic pressure transmitted to two wheel cylinders **40** provided on the vehicle wheels RR, RL, FR, and FL, a first cut valve **261** provided on the first backup flow path **251** connecting the first hydraulic port **24a** and the first hydraulic circuit **201** to control the flow of the hydraulic pressure, a second cut valve **262** provided on a second backup flow path **252** connecting the second hydraulic port **24b** and the second hydraulic circuit **202** to control the flow of the hydraulic pressure, and an electronic control unit for controlling the hydraulic supply device **100** and valves **54**, **60**, **221a**, **221b**, **221c**, **221d**, **222a**, **222b**, **222c**, **222d**, **233**, **235**, **236**, and **243** based on hydraulic pressure information and pedal displacement information.

(31) The electronic control unit may perform the overall control of the electromechanical brake system **1**.

(32) The electronic control unit **2000** may perform the overall control of the electromechanical brake system **1**.

(33) The hydraulic provision unit **110** includes a cylinder block **111** in which a pressure chamber for receiving and storing a brake fluid is formed, a hydraulic piston **114** accommodated in the cylinder block **111**, a hydraulic piston **114** accommodated in the cylinder block **111**, a sealing member **115** (**115a** and **115b**) provided between the hydraulic piston **114** and the cylinder block **111** to seal a pressure chamber, and a driving shaft **133** connected to a rear end of the hydraulic piston **114** to transmit power output from the power converter **130** to the hydraulic piston **114**.

(34) The pressure chamber may include a first pressure chamber **112** positioned in front of the hydraulic piston **114** (forward direction or left direction in the drawing) and a second pressure chamber **113** positioned behind the hydraulic piston **114** (reverse direction or right direction in the drawing). In other words, the first pressure chamber **112** is partitioned by the cylinder block **111** and a front end of the hydraulic piston **114** and provided so that a volume varies according to the

movement of the hydraulic piston **114**, and the second pressure chamber **113** is partitioned by the cylinder block **111** and the rear end of the hydraulic piston **114** and provided so that a volume varies according to the movement of the hydraulic piston **114**.

(35) The first and second pressure chambers **112** and **113** may be connected to the reservoir **30** by dump flow paths **116** and **117**, respectively, and may receive and store a brake fluid from the reservoir **30** or transmit the brake fluid in the first or second pressure chamber **112** or **113** to the reservoir **30**.

(36) Next, flow paths **211**, **212**, **213**, **214**, **215**, **216**, and **217** and valves **231**, **232**, **233**, **234**, **235**, **236**, **241**, **242**, and **243** connected to the first pressure chamber **112** and the second pressure chamber **113** will be described.

(37) The second hydraulic flow path **212** may communicate with the first hydraulic circuit **201**, and the third hydraulic flow path **213** may communicate with the second hydraulic circuit **202**.

Therefore, hydraulic pressure may be transmitted to the first hydraulic circuit **201** and the second hydraulic circuit **202** by the advance of the hydraulic piston **114**.

(38) In addition, the electromechanical brake system **1** may include a first control valve **231** and a second control valve **232** respectively provided on the second and third hydraulic flow paths **212** and **213** to control the flow of the brake fluid.

(39) In addition, the first and second control valves **231** and **232** may be provided as a check valve for allowing only the flow of the brake fluid in a direction from the first pressure chamber **112** toward the first or second hydraulic circuit **201** or **202** and blocking the flow of the brake fluid in the opposite direction. In other words, the first or second control valve **231** or **232** may allow the hydraulic pressure of the first pressure chamber **112** to be transmitted to the first or second hydraulic circuit **201** or **202** and prevent the hydraulic pressure of the first or second hydraulic circuit **201** or **202** from leaking to the first pressure chamber **112** through the second or third hydraulic flow path **212** or **213**.

(40) Meanwhile, the fourth hydraulic flow path **214** may be branched off into a fifth hydraulic flow path **215** and a sixth hydraulic flow path **216** and may communicate with both of the first hydraulic circuit **201** and the second hydraulic circuit **202**. For example, the fifth hydraulic flow path **215** branched off from the fourth hydraulic flow path **214** may communicate with the first hydraulic circuit **201**, and the sixth hydraulic flow path **216** branched off from the fourth hydraulic flow path **214** may communicate with the second hydraulic circuit **202**. Therefore, hydraulic pressure may be transmitted to both of the first hydraulic circuit **201** and the second hydraulic circuit **202** by retreat of the hydraulic piston **114**.

(41) In addition, the electromechanical brake system **1** may include a third control valve **233** provided on the fifth hydraulic flow path **215** to control the flow of the brake fluid and a fourth control valve **234** provided on the sixth hydraulic flow path **216** to control the flow of the brake fluid.

(42) The third control valve **233** may be provided as a bi-directional control valve for controlling the flow of the brake fluid between the second pressure chamber **113** and the first hydraulic circuit **201**. In addition, the third control valve **233** may be provided as a normally closed type solenoid valve operated so that the valve is opened when receiving an opening signal from the electronic control unit in a normally closed state.

(43) In addition, the fourth control valve **234** may be provided as a check valve for allowing only the flow of the brake fluid in a direction from the second pressure chamber **113** to the second hydraulic circuit **202** and blocking the flow of the brake fluid in an opposite direction. In other words, the fourth control valve **234** can prevent the hydraulic pressure of the second hydraulic circuit **202** from leaking to the second pressure chamber **113** through the sixth hydraulic flow path **216** and the fourth hydraulic flow path **214**.

(44) In addition, the electromechanical brake system **1** may include a fifth control valve **235** provided on the seventh hydraulic flow path **217** connecting the second hydraulic flow path **212**

and the third hydraulic flow path **213** to control the flow of the brake fluid and a sixth control valve **236** provided on the eighth hydraulic flow path **218** connecting the second hydraulic flow path **212** and the seventh hydraulic flow path **217** to control the flow of the brake fluid. In addition, the fifth control valve **235** and the sixth control valve **236** may be provided as normally closed type solenoid valves operated so that the valves are opened when receiving the opening signal from the electronic control unit in the normally closed state.

(45) The fifth control valve **235** and the sixth control valve **236** may be operated to be opened when an abnormality occurs in the first control valve **231** or the second control valve **232** so that the hydraulic pressure in the first pressure chamber **112** may be transmitted to both of the first hydraulic circuit **201** and the second hydraulic circuit **202**.

(46) In addition, the fifth control valve **235** and the sixth control valve **236** may be operated to be opened when the hydraulic pressure in the wheel cylinder **40** is suctioned and sent to the first pressure chamber **112**. This is because the first control valve **231** and the second control valve **232** provided on the second hydraulic flow path **212** and the third hydraulic flow path **213** are provided as the check valves for allowing only the unidirectional flow of the hydraulic fluid.

(47) In addition, the electromechanical brake system **1** may further include a first dump valve **241** and a second dump valve **242** respectively provided on the first and second dump flow paths **116** and **117** to control the flow of the brake fluid. The dump valves **241** and **242** may be check valves opened only in a direction from the reservoir **30** to the first or second pressure chamber **112** or **113** and closed in an opposite direction. In other words, the first dump valve **241** may be the check valve for allowing the flow of the brake fluid from the reservoir **30** to the first pressure chamber **112** and blocking the flow of the brake fluid from the first pressure chamber **112** to the reservoir **30**, and the second dump valve **242** may be the check valve for allowing the flow of the brake fluid from the reservoir **30** to the second pressure chamber **113** and blocking the flow of the brake fluid from the second pressure chamber **113** to the reservoir **30**.

(48) In addition, the second dump flow path **117** may include a bypass flow path, and the third dump valve **243** for controlling the flow of the brake fluid between the second pressure chamber **113** and the reservoir **30** may be installed on the bypass flow path.

(49) The third dump valve **243** may be provided as the solenoid valve capable of controlling the bi-directional flow and provided as a normally open type solenoid valve operated so that the valve is closed when receiving a closing signal from the electronic control unit in a normally open state.

(50) The hydraulic provision unit **110** of the electromechanical brake system **1** may be operated in a double-acting manner. In other words, as the hydraulic piston **114** advances, the hydraulic pressure generated in the first pressure chamber **112** may be transmitted to the first hydraulic circuit **201** through the first hydraulic flow path **211** and the second hydraulic flow path **212** to operate the wheel cylinders **40** installed on the front right wheel FR and the rear left wheel RL and transmitted to the second hydraulic circuit **202** through the first hydraulic flow path **211** and the third hydraulic flow path **213** to operate the wheel cylinders **40** installed on the rear right wheel RR and the front left wheel FL.

(51) Likewise, the hydraulic pressure generated in the second pressure chamber **113** as the hydraulic piston **114** retreats may be transmitted to the first hydraulic circuit **201** through the fourth hydraulic flow path **214** and the fifth hydraulic flow path **215** to operate the wheel cylinders **40** installed on the front right wheel FR and the rear left wheel RL and transmitted to the second hydraulic circuit **202** through the fourth hydraulic flow path **214** and the sixth hydraulic flow path **216** to operate the wheel cylinders **40** installed on the rear right wheel RR and the front left wheel FL.

(52) In addition, a negative pressure generated in the first pressure chamber **112** as the hydraulic piston **114** retreats may suction the brake fluid in the wheel cylinders **40** installed on the front right wheel FR and the rear left wheel RL to transmit the brake fluid to the first pressure chamber **112** through the first hydraulic circuit **201**, the second hydraulic flow path **212**, and the first hydraulic

flow path **211** and suction the brake fluid in the wheel cylinders **40** installed on the rear right wheel RR and the front left wheel FL to transmit the brake fluid to the first pressure chamber **112** through the second hydraulic circuit **202**, the third hydraulic flow path **213**, and the first hydraulic flow path **211**.

(53) Next, the motor **120** and the power converter **130** of the hydraulic supply device **100** will be described.

(54) The motor **120** is a device for generating a rotational force by a signal output from the electronic control unit and may generate the rotational force in a forward or reverse direction. A rotational angular velocity and a rotational angle of the motor **120** may be precisely controlled.

(55) In addition, a motor position sensor MPS is a motor control sensor for controlling the rotational angle or a current of the motor **120**.

(56) In other words, the hydraulic supply device **100** may further include the motor position sensor MPS for measuring the rotational angular velocity and rotational angle of the motor **120** and send information on rotational angles and positions of the motor **120** to the electronic control unit.

(57) Meanwhile, the electronic control unit includes the motor **120** and controls the plurality of valves **54**, **60**, **221a**, **221b**, **221c**, **221d**, **222a**, **222b**, **222c**, **222d**, **233**, **235**, **236**, and **243**.

(58) In addition, an operation of controlling the plurality of valves according to the displacement of the brake pedal **10** in the electronic control unit will be described below.

(59) A driving force of the motor **120** generates the displacement of the hydraulic piston **114** through the power converter **130**, and the hydraulic pressure generated as the hydraulic piston **114** slides in the pressure chamber is transmitted to the wheel cylinders **40** installed on the respective vehicle wheels RR, RL, FR, and FL through the first and second hydraulic flow paths **211** and **212**.

(60) The power converter **130** is a device for converting the rotational force into a linear motion and may include, for example, a worm shaft **131**, a worm wheel **132**, and the driving shaft **133**.

(61) In other words, a displacement signal detected by the pedal position sensor **11** when the displacement of the brake pedal **10** occurs is transmitted to the electronic control unit, and the electronic control unit drives the motor **120** in one direction and rotates the worm shaft **131** in the one direction. The rotational force of the worm shaft **131** is transmitted to the driving shaft **133** through the worm wheel **132**, and the hydraulic piston **114** connected to the driving shaft **133** moves forward and generates hydraulic pressure in the first pressure chamber **112**.

(62) Conversely, when the pressing force is released from the brake pedal **10**, the electronic control unit drives the motor **120** in an opposite direction so that the worm shaft **131** rotates in the opposite direction. Therefore, the worm wheel **132** also rotates in the opposite direction, and as the hydraulic piston **114** connected to the driving shaft **133** returns (retreats), a negative pressure is generated in the first pressure chamber **112**.

(63) Meanwhile, the hydraulic pressure and the negative pressure may also be generated in a direction opposite to the above direction. In other words, the displacement signal detected by the pedal position sensor **11** when the displacement of the brake pedal **10** occurs is transmitted to the electronic control unit, and the electronic control unit drives the motor **120** in the opposite direction and rotates the worm shaft **131** in the opposite direction. The rotational force of the worm shaft **131** is transmitted to the driving shaft **133** through the worm wheel **132**, and hydraulic pressure is generated in the second pressure chamber **113** as the hydraulic piston **114** connected to the driving shaft **133** retreats.

(64) Conversely, when the pressing force is removed from the brake pedal **10**, the electronic control unit drives the motor **120** in one direction so that the worm shaft **131** rotates in the one direction. Therefore, the worm wheel **132** also rotates in the opposite direction, and as the hydraulic piston **114** connected to the driving shaft **133** returns (advances), a negative pressure is generated in the second pressure chamber **113**.

(65) As described above, the hydraulic supply device **100** serves to transmit the hydraulic pressure to the wheel cylinder **40** or suction the hydraulic pressure according to the rotational direction of

the rotational force generated from the motor **120** to transmit the hydraulic pressure to the reservoir **30**.

(66) Meanwhile, when the motor **120** rotates in one direction, hydraulic pressure may be generated in the first pressure chamber **112** or a negative pressure may be generated in the second pressure chamber **113**, and whether braking is performed using the hydraulic pressure or is released using the negative pressure may be determined by controlling the valves **54**, **60**, **221a**, **221b**, **221c**, **221d**, **222a**, **222b**, **222c**, **222d**, **233**, **235**, **236**, and **243** constituting a plurality of valves **800**.

(67) In addition to a structure of a ball-screw-nut assembly, it should be understood that any structure of the power converter **130** according to one embodiment may be adopted as long as the structure can convert a rotational motion into a linear motion.

(68) The first cut valve **261** for controlling the flow of the brake fluid may be provided on the first backup flow path **251**, and a second cut valve **262** for controlling the flow of the brake fluid may be provided on the second backup flow path **252**. In addition, the first backup flow path **251** may connect the first hydraulic port **24a** and the first hydraulic circuit **201**, and the second backup flow path **252** may connect the second hydraulic port **24b** and the second hydraulic circuit **202**.

(69) In addition, the first and second cut valves **261** and **262** may be normally open type solenoid valves operated so that the valves are closed when receiving the closing signal from the electronic control unit in a normally open state.

(70) Next, referring to FIG. **1**, the hydraulic control unit **200** according to one embodiment will be described.

(71) The hydraulic control unit **200** may include the first hydraulic circuit **201** and the second hydraulic circuit **202** for receiving hydraulic pressure, each of which controls two wheels. For example, the first hydraulic circuit **201** may control the front right wheel FR and the rear left wheel RL, and the second hydraulic circuit **202** may control the front left wheel FL and the rear right wheel RR. In addition, the wheel cylinder **40** is installed on each of the vehicle wheels FR, FL, RR, and RL to receive hydraulic pressure and perform braking.

(72) The first hydraulic circuit **201** is connected to the first hydraulic flow path **211** and the second hydraulic flow path **212** to receive hydraulic pressure from the hydraulic supply device **100** or the master cylinder **20**, and the second hydraulic flow path **212** is branched off into two flow paths connected to the front right wheel FR and the rear left wheel RL. Likewise, the second hydraulic circuit **202** is connected to the first hydraulic flow path **211** and the third hydraulic flow path **213** to receive hydraulic pressure from the hydraulic supply device **100**, and the third hydraulic flow path **213** is branched off into two flow paths connected to the front left wheel FL and the rear right wheel RR.

(73) The hydraulic circuits **201** and **202** may include a plurality of inlet valves **221** (**221a**, **221b**, **221c**, and **221d**) to control the flow of the hydraulic pressure. For example, the first hydraulic circuit **201** may be provided with the two inlet valves **221a** and **221b** connected to the first hydraulic flow path **211** to control the hydraulic pressure transmitted to the two wheel cylinders **40**. In addition, the second hydraulic circuit **202** may be provided with the two inlet valves **221c** and **221d** connected to the second hydraulic flow path **212** to control the hydraulic pressure transmitted to the wheel cylinders **40**.

(74) In addition, the inlet valve **221** may be provided as the normal open type solenoid valve disposed upstream of the wheel cylinder **40** and operated so that the valve is closed when receiving the closing signal from the electronic control unit in the normally open state.

(75) In addition, the hydraulic circuits **201** and **202** may include check valves **223a**, **223b**, **223c**, and **223d** provided on bypass flow paths connecting the fronts and rears of the respective inlet valves **221a**, **221b**, **221c**, and **221d**. The check valves **223a**, **223b**, **223c**, and **223d** may be provided to allow only the flow of the brake fluid in a direction from the wheel cylinder **40** toward the hydraulic provision unit **110** and limit the flow of the brake fluid in a direction from the hydraulic provision unit **110** toward the wheel cylinder **40**. The check valves **223a**, **223b**, **223c**, and **223d**

may quickly discharge the braking pressure of the wheel cylinder **40**, and when the inlet valves **221a**, **221b**, **221c**, and **221d** do not operate normally, the hydraulic pressure of the wheel cylinder **40** may be introduced into the hydraulic provision unit **110**.

(76) In addition, the hydraulic circuits **201** and **202** may further include a plurality of outlet valves **222** (**222a**, **222b**, **222c**, and **222d**) connected to the reservoir **30** to improve performance when braking is released. Each of the outlet valves **222** is connected to the wheel cylinder **40** to control the hydraulic pressure discharged from each of the vehicle wheels RR, RL, FR, and FL. In other words, the outlet valve **222** may detect the braking pressure of each of the vehicle wheels RR, RL, FR, and FL and may be selectively opened to control the pressure when braking pressure reduction is required.

(77) In addition, the outlet valve **222** may be provided as the normally closed type solenoid valve operated so that the valve is opened when receiving the opening signal from the electronic control unit in the normally closed state.

(78) In addition, the hydraulic control unit **200** may be connected to the backup flow paths **251** and **252**. For example, the first hydraulic circuit **201** may be connected to the first backup flow path **251** to receive hydraulic pressure from the master cylinder **20**, and the second hydraulic circuit **202** may be connected to the second backup flow path **252** to receive hydraulic pressure from the master cylinder **20**.

(79) At this time, the first backup flow path **251** may join the first hydraulic circuit **201** upstream of the first and second inlet valves **221a** and **221b**. Likewise, the second backup flow path **252** may join the second hydraulic circuit **202** upstream of the third and fourth inlet valves **221c** and **221d**. Therefore, when the first and second cut valves **261** and **262** are closed, the hydraulic pressure provided from the hydraulic supply device **100** may be supplied to the wheel cylinders **40** through the first and second hydraulic circuits **201** and **202**, and when the first and second cut valves **261** and **262** are opened, the hydraulic pressure provided from the master cylinder **20** may be supplied to the wheel cylinders **40** through the first and second backup flow paths **251** and **252**. At this time, since the plurality of inlet valves **221a**, **221b**, **221c**, and **221d** are in open states, there is no need to change operation states.

(80) Referring to FIG. 2, the electromechanical brake system **1** according to one embodiment of the present disclosure needs to determine a section in which a required fluid amount of brake oil of a vehicle may be monitored. The required fluid amount of the brake oil represents the relationship between the position of the motor and the brake fluid pressure (pressure). Therefore, since the required fluid amount of the brake oil may vary depending on the state of the vehicle, the required fluid amount may temporarily vary, and thus it is necessary to exclude a case in which the required fluid amount varies temporarily. In other words, the required fluid amount of the brake oil may represent a volume of the brake oil required by the motor according to pressure.

(81) To this end, a controller **220** may check the section in which the required fluid amount of the brake oil may be monitored through a temperature sensor **212**, a pedal displacement sensor **214**, and a pressure sensor **216**. For example, the controller **220** monitors the required fluid amount of the brake oil through the temperature sensor **212** when an estimated brake temperature is lower than a set temperature and an outside air temperature is higher than the set temperature. In other words, the controller **220** monitors the required fluid amount of the brake oil at temperatures excluding a high estimated brake temperature and a low outer air temperature.

(82) To this end, a controller **220** may check the section in which the required fluid amount of the brake oil may be monitored through a temperature sensor **2121**, a pedal displacement sensor **2123**, and a pressure sensor **2125**. For example, the controller **220** monitors the required fluid amount of the brake oil through the temperature sensor **2121** when an estimated brake temperature is lower than a set temperature and an outside air temperature is higher than the set temperature. In other words, the controller **220** monitors the required fluid amount of the brake oil at temperatures excluding a high estimated brake temperature and a low outer air temperature.

(83) Therefore, the controller **220** monitors the required fluid amount when the required fluid amount that is increased with respect to the required fluid amount reference, which is a basic required fluid amount according to a map, satisfies a predetermined condition. Here, the required fluid amount reference according to the map is stored in a storage **230** of the electromechanical brake system **1**.

(84) In addition, the controller **220** calculates a ratio of the required fluid amount having an increased required fluid amount and the required fluid amount reference according to the map. To this end, the controller **220** uses a position of the motor and an area of the piston in order to first calculate the required fluid amount. In other words, the required fluid amount may be calculated by multiplying the position of the motor by the area of the piston.

(85) The controller **220** calculates the ratio of the required fluid amount and the required fluid amount reference using the thus calculated required fluid amount. In other words, the controller **220** calculates the ratio by dividing the required fluid amount by the required fluid amount reference.

(86) As described above, the controller **220** changes the required fluid amount reference stored in the storage **230** according to the calculated ratio of the required fluid amount and determines whether to apply the changed required fluid amount reference. In order to determine whether to apply the changed required fluid amount reference, the controller **220** checks whether the ratio has been calculated at least three times in a pressure zone of one section and checks whether the ratio has been calculated in pressure zones of at least three sections. In addition, the controller **220** checks whether a traveling distance of the vehicle is a predetermined distance or more and checks whether the traveling distance is the predetermined distance or more after the required fluid amount reference has been corrected.

(87) Only when the above condition is satisfied, the controller **220** applies the calculated required fluid amount to the required fluid amount reference previously stored in the storage **230** to change the required fluid amount reference previously stored.

(88) Referring to FIG. 5, a method of controlling the electromechanical brake system **1** according to one embodiment of the present disclosure will be described. The method of controlling the electromechanical brake system **1** according to one embodiment of the present disclosure will be described with reference to FIGS. **1** to **4**.

(89) Whether monitoring is possible is determined (S**101**).

(90) In the electromechanical brake system **1**, brake oil is required to generate a braking force by the motor. A required fluid amount is actually changed depending on a state of a vehicle as compared to a required fluid amount reference according to a map, and the controller **220** needs to monitor how much the required fluid amount is changed as compared to the required fluid amount reference.

(91) In this operation, as described above, whether to monitor the required fluid amount is determined. Here, since there is a case in which the required fluid amount is temporarily changed in the electromechanical brake system **1**, the controller **220** monitors the continuously changing required fluid amount except for a case in which the required fluid amount is temporarily changed.

(92) The controller **220** determines that a case in which an estimated brake temperature is higher than a set temperature and a case in which an outside air temperature is lower than the set temperature are cases in which the required fluid amount is temporarily changed. In addition, the controller **220** determines that a case in which the required fluid amount is actively increased by a driver, a case in which the vehicle suddenly decelerates, and a case in which the brake oil pressure is low are cases in which the required fluid amount is temporarily changed.

(93) Therefore, in this operation, as described above, the controller **220** determines that cases except for the case in which the required fluid amount is temporarily changed are in a situation in which the required fluid amount may be monitored.

(94) When it is determined that monitoring of the required fluid amount is possible, the required

fluid amount according to pressure is calculated (S103).

(95) The required fluid amount according to pressure is calculated using a position of the motor and an area of the piston. In this operation, the controller **220** calculates the required fluid amount in real time according to pressure as shown in FIG. 3 in order to calculate the required fluid amount.

(96) As described above, when the required fluid amount is calculated, the ratio of the calculated required fluid amount and the required fluid amount reference is calculated (S105).

(97) The controller **220** calculates the ratio of the required fluid amount and the required fluid amount reference. The ratio of the required fluid amount and the required fluid amount reference may be an increase in the required fluid amount with respect to the required fluid amount reference at the same pressure.

(98) In this operation, as shown in FIG. 4, the ratio of the required fluid amount and the required fluid amount reference is divided into a plurality of sections according to pressure, calculated in at least three sections among a plurality of sections, and also calculated three times or more in one section.

(99) The controller **220** calculates the ratio of the required fluid amount and the required fluid amount reference multiple times in the plurality of sections as described above.

(100) As described above, when the ratio of the required fluid amount and the required fluid amount reference is calculated, whether the calculated ratio is applicable is determined (S107).

(101) With regard to whether the calculated ratio is applicable, the controller **220** determines that the calculated ratio is applicable in a case in which the ratio has been calculated multiple times in the plurality of sections in the operation S105. In addition, the controller **220** determines that the calculated ratio is applicable when a traveling distance is a predetermined distance or more after the vehicle is mass-produced and determines that the calculated ratio is applicable when the traveling distance is the predetermined distance or more after the required fluid amount reference has been corrected.

(102) Here, “applicable” means that the controller **220** changes the required fluid amount reference stored in the storage **230** to the required fluid amount calculated in the operation S103 and stores the calculated required fluid amount.

(103) Here, when the calculated required fluid amount may not be applied for replacing the current required fluid amount reference, the method may return to the operation S101.

(104) When it is determined that the required fluid amount is applicable, the fluid required amount of the motor is applied as the required fluid amount reference (S109).

(105) When it is determined in the operation S107 that the required fluid amount calculated in the operation S105 is applicable, the controller **220** stores the required fluid amount calculated in the operation S105 in the storage **230** instead of the required fluid amount reference stored in the storage **230**. In the operation, the required fluid amount newly stored is used as the required fluid amount reference later.

(106) As is apparent from the above description, it is possible to monitor the variation of the required fluid amount of brake oil according to a state of a vehicle in real time and apply a monitored result to a required fluid amount reference corresponding to a reference amount, thereby preventing the degradation of brake performance.

(107) Exemplary embodiments of the present disclosure have been described above. In the exemplary embodiments described above, some components may be implemented as a “module”. Here, the term ‘module’ means, but is not limited to, a software and/or hardware component, such as a Field Programmable Gate Array (FPGA) or Application Specific Integrated Circuit (ASIC), which performs certain tasks. A module may advantageously be configured to reside on the addressable storage medium and configured to execute on one or more processors.

(108) Thus, a module may include, by way of example, components, such as software components, object-oriented software components, class components and task components, processes, functions, attributes, procedures, subroutines, segments of program code, drivers, firmware, microcode,

circuitry, data, databases, data structures, tables, arrays, and variables. The operations provided for in the components and modules may be combined into fewer components and modules or further separated into additional components and modules. In addition, the components and modules may be implemented such that they execute one or more CPUs in a device.

(109) With that being said, and in addition to the above described exemplary embodiments, embodiments can thus be implemented through computer readable code/instructions in/on a medium, e.g., a computer readable medium, to control at least one processing element to implement any above described exemplary embodiment. The medium can correspond to any medium/media permitting the storing and/or transmission of the computer readable code.

(110) The computer-readable code can be recorded on a medium or transmitted through the Internet. The medium may include Read Only Memory (ROM), Random Access Memory (RAM), Compact Disk-Read Only Memories (CD-ROMs), magnetic tapes, floppy disks, and optical recording medium. Also, the medium may be a non-transitory computer-readable medium. The media may also be a distributed network, so that the computer readable code is stored or transferred and executed in a distributed fashion. Still further, as only an example, the processing element could include at least one processor or at least one computer processor, and processing elements may be distributed and/or included in a single device.

(111) While exemplary embodiments have been described with respect to a limited number of embodiments, those skilled in the art, having the benefit of this disclosure, will appreciate that other embodiments can be devised which do not depart from the scope as disclosed herein. Accordingly, the scope should be limited only by the attached claims.

Claims

1. An electromechanical brake system comprising: a temperature sensor configured to measure one or more of a brake oil temperature and an outside air temperature; a pressure sensor configured to measure a pressure for a required fluid amount depending on an actual pressure for brake oil required to generate a braking force by a motor; and a controller configured to determine whether to monitor the required fluid amount based on the pressure for the required fluid amount, and one or more of a brake temperature estimated by the temperature sensor and the outside air temperature, and in response to monitoring of the required fluid amount, calculate a ratio of the required fluid amount and a required fluid amount reference according to a map and the required fluid amount, wherein the controller is configured to monitor the required fluid amount when the brake temperature is lower than a set first temperature and the outside air temperature is higher than a set second temperature.
2. The electromechanical brake system of claim 1, wherein the controller applies the required fluid amount as the required fluid amount reference according to the ratio of the required fluid amount and the required fluid amount reference according to the map.
3. The electromechanical brake system of claim 1, wherein the controller monitors the required fluid amount in cases in which the required fluid amount is continuously changed.
4. The electromechanical brake system of claim 3, wherein the cases in which the required fluid amount is continuously changed are cases excluding a case in which the brake temperature is higher than the set first temperature and a case in which the outside air temperature is lower than the set second temperature.
5. The electromechanical brake system of claim 3, wherein the cases in which the required fluid amount is continuously changed are cases excluding one or more among a case in which the required fluid amount is actively increased by a driver, a case in which a vehicle suddenly decelerates, and a case in which a brake oil pressure is low.
6. The electromechanical brake system of claim 1, wherein the controller determines whether the required fluid amount is applicable as the required fluid amount reference, and applies the required

fluid amount as the required fluid amount reference.

7. The electromechanical brake system of claim 6, wherein the controller determines that the ratio of the required fluid amount and the required fluid amount reference is applicable when calculated in a plurality of sections divided according to the pressure.

8. The electromechanical brake system of claim 6, wherein the controller determines that the ratio of the required fluid amount and the required fluid amount reference is applicable when calculated multiple times in one or more among a plurality of sections divided according to the pressure.

9. The electromechanical brake system of claim 6, wherein the controller determines that the ratio of the required fluid amount and the required fluid amount reference is applicable when a traveling distance of a vehicle to which the braking force is provided by the motor is a predetermined distance or more.

10. The electromechanical brake system of claim 6, wherein the controller determines that the ratio is applicable when a traveling distance of a vehicle to which the braking force is provided by the motor is a predetermined distance or more after the required fluid amount has been applied as the required fluid amount reference.
